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Rates of knee arthroplasty after genicular nerve ablation: A retrospective study utilizing a large national database

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ABSTRACT

Background: Genicular nerve ablation (GNA), which includes radiofrequency ablation (GNRFA), cryoneurolysis, and chemical neurolysis, is a minimally invasive procedure with growing evidence for improved pain and functional outcomes in the treatment of knee osteoarthritis (KOA). With conservative management for KOA having variable efficacy and knee arthroplasty (KA) carrying substantial risks, there is utility in identifying if GNA may delay or defer surgery.

Objective: The purpose of this study was to assess the rate of patients receiving KA, including total knee arthroplasty (TKA) or unicompartmental knee arthroplasty (UKA) after receiving GNA.

Methods: TriNetX, a national database, was queried for patients 18 years or older who underwent GNA between 2004 and 2025. Patients with severe medical comorbidities and patients with prior TKA or UKA were excluded. Descriptive analyses were performed using the TriNetX Analytics platform and the incidence of post-GNA TKA or UKA was calculated. Secondary analysis calculating the cumulative incidence of arthroplasty among patients without prior KA within 3 months, 6 months, 1 year, 2 years, and 5 years after GNA was also assessed.

Results: There were 6035 patients who underwent GNA during this study period and who were included in the final analysis after excluding patients with select comorbidities or who had a prior TKA or UKA. Among this cohort, 795 (13.2 %; 95 % CI 12.3–14.0) patients underwent TKA (N = 753; 12.5 %; 95 % CI 11.7–13.3) or UKA (N = 42; 0.7 %; 95 % CI 0.5–0.9) after receiving GNA. For patients without prior KA, 152 patients underwent KA at 3 months post-GNA (1.3 %; 95 % CI 1.1–1.5), 415 patients underwent KA at 6 months post-GNA (3.5 %; 95 % CI 3.2–3.9), 847 patients underwent KA at 1 year post-GNA (7.2 %; 95 % CI 6.7–7.6), 1219 patients underwent KA at 2 years post-GNA (10.3 %; 95 % CI 9.8–10.9), and 1469 patients underwent KA at 5 years post-GNA (12.4 %; 95 % CI 11.8–13.0).

Conclusion: This is the largest cohort study to identify rates of KA following GNA. We found that roughly one in eight patients in this cohort received TKA or UKA after GNA, suggesting that GNA may offer clinically meaningful symptom relief for patients with KOA, such that KA may be delayed or deferred. Causality for whether GNA delays or prevents KA cannot be established from this study. The cost effectiveness and surgical sparing efficacy of GNA for symptomatic KOA should be further explored.

1. Introduction

Knee osteoarthritis (KOA) is a leading cause of pain, functional limitation, and disability in the world, with a global prevalence of about 23 % in adults over 40 years old and lifetime risk of symptomatic KOA of 45 % by age 85 [1,2]. Conservative management of KOA includes exercise, weight loss, physical therapy, non-steroidal anti-inflammatory drugs (NSAIDs), as well as intraarticular injections, such as steroids, hyaluronic acid (HA), and/or platelet rich plasma (PRP) [3]. However, NSAIDs come with adverse effects and an FDA black box warning. Intra-articular steroid injections have short-lived efficacy and can have negative effects on cartilage. HA and PRP have been shown to have variable outcomes, with the latter lacking standardized protocols and formulations [4]. Where conservative treatments fail, total knee arthroplasty (TKA) or unicompartmental knee arthroplasty (UKA) may be considered [5]. However, these surgeries carry significant risks, such as infection, thromboembolism, and perioperative mortality. Furthermore, patients may either be poor surgical candidates due to medical

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comorbidities or wish to avoid surgery due to personal preference [6–9].

Genicular nerve ablation (GNA), which encompasses radiofrequency ablation (GNRFA), cryoneurolysis, and chemical neurolysis, has emerged as a minimally invasive option for patients with chronic pain due to knee OA. These procedures involve sensory denervation of the articular sensory branches of the knee joint capsule [10,11]. GNRFA, which includes thermal, cooled, or pulsed techniques, is the most common of the GNA procedures and has been shown to have significant positive effects on pain, functional outcomes, and patient reported outcome scores across randomized controlled trials (RCT), systematic reviews, and real-world cohort studies. Benefits have been shown to last 6–12 months or longer [10–16]. Several pragmatic studies have also compared cooled GNRFA to common intraarticular injectates for KOA, such as corticosteroid and HA, and demonstrated significant reduction in pain scores [16–20]. Regarding cryoneurolysis, a single RCT by Radnovich et al. demonstrated significant improvements in pain and functional outcome scores when compared to sham at 30- and 90-days post-procedure [21]. While there are ongoing registered RCTs on cryoneurolysis for KOA pain, one of which compares it to cooled GNRFA, it should be noted that most of the current research discusses pre-operative cryoneurolysis for post-TKA pain [22–24]. Chemical neurolysis, typically performed with phenol or alcohol, has been less studied. A single RCT by Elashmawy et al. demonstrated significant improvements in pain and functional outcomes scores in favor of phenol compared to steroid for GNA, particularly at 6 months [25]. A prospective study by Risso et al. demonstrated similar positive outcomes at 6 months [26].

Despite existing and ongoing research, little is known about the whether GNA influences the long-term need for surgical intervention. More specifically, the extent to which GNA may delay or reduce conversion to knee arthroplasty (KA) remains largely unquantified. Only two of the previously referenced studies report data on patients who went on to receive KA post-GNA [19,21]. Their sample sizes are small. The purpose of this study therefore was to estimate the rate of TKA or UKA after undergoing GNA utilizing a large national database.

2. Methods

2.1. Study design

This study utilized de-identified patient data obtained through the TriNetX Research Network, a global federated database providing aggregated electronic health record information from participating healthcare organizations. All data available within TriNetX are de-identified in accordance with the Health Insurance Portability and Accountability Act (HIPAA) and other relevant privacy regulations. No protected health information was accessed or disclosed during this study.

Our Institutional Review Board determined that analyses conducted using de-identified data from TriNetX do not meet the definition of human subjects research and are therefore exempt from IRB review. All study procedures were performed in accordance with institutional policies, applicable national research ethics standards, and the principles outlined in the Declaration of Helsinki. Informed consent was not required given the use of de-identified, aggregated data.

The TriNetX Research Network is a federated database that compiles de-identified electronic health records from more than 130 million patients across over 100 healthcare organizations in the United States. Data can be accessed and analyzed through queries utilizing International Classification of Diseases, Tenth Revision (ICD-10) diagnostic codes and Current Procedural Terminology (CPT) procedural codes. For this study, TriNetX was queried to identify all adult patients (≥18 years) who underwent GNA (CPT 64624) in the United States between January 2024 and September 2025. From this cohort, the number of patients that went on to have a TKA (CPT 27447) or UKA (CPT 27446) after GNA was extracted. Patients were then excluded if they had ICD-10 diagnosis of morbid obesity (E66.01), uncontrolled diabetes (E11.65), COPD (J44), heart failure (I50) and/or ESRD (N18.6) to minimize confounding from patients whose medical comorbidities may preclude them from receiving a TKA or UKA. Patients were also excluded if they underwent TKA or UKA prior to GNA. Patients lacking either pre-procedure data or any follow-up data after GNA were excluded. As the database cannot restrict a minimum follow-up duration, patients were only included if their GNA procedure occurred at least one year before the query period. Demographic variables including age, sex, race, and ethnicity were also collected. The primary outcome was the incidence of TKA or UKA after GNA. Secondary outcomes included the cumulative proportion of patients undergoing TKA or UKA within 3 months, 6 months, 1 year, 2 years, and 5 years after GNA.

2.2. Statistical analysis

All analyses were performed using the TriNetX Analytics platform (Cambridge, MA, USA). Descriptive statistics were used to summarize patient demographics and procedural outcomes. Continuous variables were reported as means ± standard deviations (SD), and categorical variables as frequencies and percentages.

The primary outcome was the incidence of post-GNA TKA or UKA during the follow-up period. For the full GNA cohort and for the subcohorts excluding (1) patients with prior TKA or UKA and (2) patients with severe comorbidities, the proportion of patients who underwent TKA or UKA after GNA was calculated. For each proportion, 95 % confidence intervals (CIs) were estimated using the Wilson score method.

For secondary outcomes, we calculated the incidence of arthroplasty over time among patients without prior KA by identifying those who underwent TKA or UKA within 3 months, 6 months, 1 year, 2 years, and 5 years after GNA. For each time point, the cumulative incidence was defined as the number of patients with KA by that time divided by the fixed number of patients in the post-GNA subcohort that excluded patients with prior TKA or UKA. Corresponding 95 % Wilson CIs were calculated.

No inferential hypothesis testing or multivariable modeling was conducted, as the primary objective of this study was to provide descriptive estimates of post-GNA arthroplasty rates in real-world clinical practice. All confidence intervals were two-sided with a nominal coverage of 95 %.

3. Results

3.1. Baseline characteristics

A total of 13,107 patients underwent GNA during this time frame, of which 1685 (12.9 %; 95 % CI 12.3–13.4) underwent TKA (N = 1617; 12.3 %; 95 % CI 11.8–12.9) or UKA (N = 68; 0.5 %; 95 % CI 0.4–0.7) afterwards. When excluding patients with a prior TKA or UKA, 11,819 patients remained (mean age 69.7 ± 13.4 years; 68.6 % female, 31.4 % male), of which 1466 (12.4 %; 95 % CI 11.8–13.0) underwent TKA (N = 1409; 11.9 %; 95 % CI 11.3–12.5) or UKA (N = 57; 0.5 %; 95 % CI 0.4–0.6) afterwards. Additional demographic data is reported in Tables 1 and 2.

3.2. Primary outcome

A total of 6035 patients who did not have ICD-10 diagnosis of morbid obesity, uncontrolled diabetes, COPD, heart failure, and/or ESRD and CPT code for prior knee surgeries underwent GNA during this time frame. Out of this cohort, 795 (13.2 %; 95 % CI 12.3–14.0) patients underwent TKA (N = 753; 12.5 %; 95 % CI 11.7–13.3) or UKA (N = 42; 0.7 %; 95 % CI 0.5–0.9) (Table 2).

Table 1
Baseline Demographic Characteristics of Patients Undergoing Genicular Nerve Ablation (Prior Arthroplasty Excluded, n = 11,819).

Variable	Value
Age, mean ± SD, y	69.7 ± 13.4
Age at Index, mean ± SD, y	67.0 ± 13.7
Female, n (%)	8104 (68.6 %)
Male, n (%)	3715 (31.4 %)
Not Hispanic or Latino, n (%)	8458 (71.6 %)
Hispanic or Latino, n (%)	553 (4.7 %)
Unknown Ethnicity, n (%)	2808 (23.8 %)
White, n (%)	8891 (75.2 %)
Black or African American, n (%)	1740 (14.7 %)
Asian, n (%)	262 (2.2 %)
American Indian or Alaska Native, n (%)	62 (0.5 %)
Native Hawaiian or Pacific Islander, n (%)	52 (0.4 %)
Other Race, n (%)	291 (2.5 %)
Unknown Race, n (%)	521 (4.4 %)

Footnote: Values are reported as mean ± SD or number (percentage). Demographics represent patients who underwent genicular nerve ablation (CPT 64624) after exclusion of those with prior knee arthroplasty.

3.3. Secondary outcomes

Secondary outcomes were assessed in the subcohort without CPT code for prior TKA or UKA before excluding patients with severe comorbidities. At 3 months after GNA, a total of 152 patients underwent KA (1.3 %; 95 % CI 1.1–1.5), including TKA (N = 139; 1.2 %; 95 % CI 1.0–1.4) or UKA (N = 13; 0.1 %; 95 % CI 0.1–0.2). At 6 months after GNA, a total of 415 patients underwent KA (3.5 %; 95 % CI 3.2–3.9), including TKA (N = 394; 3.3 %; 95 % CI 3.0–3.7) or UKA (N = 21; 0.2 %; 95 % CI 0.1–0.3). At 1 year after GNA, a total of 847 patients underwent KA (7.2 %; 95 % CI 6.7–7.6), including TKA (N = 815; 6.9 %; 95 % CI 6.5–7.4) or UKA (N = 32; 0.3 %; 95 % CI 0.2–0.4). At 2 years after GNA, a total of 1219 patients underwent KA (10.3 %; 95 % CI 9.8–10.9), including TKA (N = 1174; 9.9 %; 95 % CI 9.4–10.5) or UKA (N = 45; 0.4 %; 95 % CI 0.3–0.5). At 5 years after GNA, a total of 1469 patients underwent KA (12.4 %; 95 % CI 11.8–13.0), including TKA (N = 1413; 12.0 %; 95 % CI 11.4–12.6) or UKA (N = 56; 0.5 %; 95 % CI 0.4–0.6) (Table 3).

4. Discussion

This retrospective database study evaluated rates of post-GNA TKA or UKA in patients without prior KA or severe medical comorbidities. These exclusion criteria were applied to avoid confounding the results with patients who may have received post-surgical GNA for pain after KA or may have been precluded from surgery, respectively. Secondary analysis evaluated cumulative incidence of post-GNA TKA or UKA at 3 months, 6 months, 1 year, 2 years, and 5 years in patients without prior KA.

4.1. Rates of knee arthroplasty

In this large, real-world analysis of more than 13,000 patients who underwent GNA, approximately one in eight subsequently received TKA or UKA. To our knowledge, this is the largest cohort study to specifically assess the rate of KA after GNA. We note that studies by Hunter et al., who studied cooled GNRFA, and Radnovich et al., who studied cryoneurolysis, reported 4 and 1 patient(s), respectively, who went on to receive KA post-GNA [19,21]. Comparative strengths of our study include its significantly larger sample size which captured 795 cases of post-GNA TKA, as well as the use of real-world data across a multitude of diverse national healthcare systems. Even after excluding patients with preexisting arthroplasty or comorbidities that typically preclude surgery, the rate of post-GNA KA remained stable at approximately 12–13 %, suggesting that many patients may derive sustained symptom relief sufficient to defer surgery during the observed follow-up period. Furthermore, time-stratified analysis demonstrated low conversion rates up to 5 years. Only 1.3 % of patients had undergone KA at 3 months, 3.5 % at 6 months, increasing to 7.2 % at 1 year, 10.3 % at 2 years, and 12.4 % at 5 years post-GNA. These cumulative rates reinforce that most patients do not proceed to KA for several years following GNA.

Prior epidemiologic data by Losina et al. suggests that 2–3 % of patients with symptomatic knee osteoarthritis proceed to knee arthroplasty annually, corresponding to approximately 15–20 % within a 3–5-year period [27]. A study by Weinstein et al. estimates that 52.2 % of males and 50.6 % of females with symptomatic KOA will receive a primary TKA before death. Among those who undergo a primary TKA, the risk of subsequent revision is 14.9 % for males and 17.4 % for females [28]. The 13.2 % conversion rate to KA observed in our primary analysis and similar conversion rates in our subsequent secondary analysis compare favorably to these numbers, though we could not calculate length of time between KOA diagnosis and GNA and/or KA.

Still, while causality cannot be inferred, this pattern supports the hypothesis that GNA may delay or reduce the need for KA, providing an additional window of improved quality of life for patients not yet ready or able to undergo surgery. The results of this study complement the growing body of evidence - previously discussed in the background of this manuscript - that GNA can offer sustained symptom improvement for patients with moderate to severe KOA who have exhausted other conservative measures.

4.2. Safety of GNA

As with most invasive procedures there are risks to GNA despite it being minimally invasive. There have been case reports of serious adverse events including full thickness skin burns [29], septic arthritis

Table 3
Cumulative incidence of knee arthroplasty over time after genicular nerve ablation (n = 11,819).

Time After GNA	Any Arthroplasty (%; 95 % CI)	TKA (%; 95 % CI)	UKA (%; 95 % CI)
3 months	1.3 % (95 % CI 1.1–1.5)	1.2 % (95 % CI 1.0–1.4)	0.1 % (95 % CI 0.1–0.2)
6 months	3.5 % (95 % CI 3.2–3.9)	3.3 % (95 % CI 3.0–3.7)	0.2 % (95 % CI 0.1–0.3)
12 months	7.2 % (95 % CI 6.7–7.6)	6.9 % (95 % CI 6.5–7.4)	0.3 % (95 % CI 0.2–0.4)
2 years	10.3 % (95 % CI 9.8–10.9)	9.9 % (95 % CI 9.4–10.5)	0.4 % (95 % CI 0.3–0.5)
5 years	12.4 % (95 % CI 11.8–13.0)	12.0 % (95 % CI 11.4–12.6)	0.5 % (95 % CI 0.4–0.6)

Footnote: Cumulative incidence values calculated with fixed denominator (n = 11,819). 95 % CIs calculated using Wilson method.

Table 2
Incidence of knee arthroplasty following genicular nerve ablation.

Outcome	Full GNA Cohort (n = 13,107)	Prior Arthroplasty Excluded (n = 11,819)	Comorbidity Excluded (n = 6035)
Any Arthroplasty	1685 (12.9; 95 % CI 12.3–13.4)	1466 (12.4; 95 % CI 11.8–13.0)	795 (13.2; 95 % CI 12.3–14.0)
TKA	1617 (12.3; 95 % CI 11.8–12.9)	1409 (11.9; 95 % CI 11.3–12.5)	753 (12.5; 95 % CI 11.7–13.3)
UKA	68 (0.5; 95 % CI 0.4–0.7)	57 (0.5; 95 % CI 0.4–0.6)	42 (0.7; 95 % CI 0.5–0.9)

Footnote: Values represent number (percentage; 95 % confidence interval), calculated using the Wilson method.

[30], hematoma formation [31], and pes anserine tendon injury [32]. While these are reported in the literature, large analyses of multiple trials demonstrate a low rate of serious adverse events. In a large systemic review of GNA written by Ajrawat et al. that included 1512 patients from 33 studies, Ajrawat reported minor adverse events that included knee swelling, subcutaneous bruising and dysesthesias after GNA, but no serious adverse events including infection, hemorrhage, thermal injury, or motor deficits [32]. This was similarly reported by Barreto et al. in a meta-analysis of 8 randomized controlled trials that included 627 patients where there were no serious adverse reports [11]. Based on these studies, GNA likely has a lower risk profile relative to KA which has several known complications including post-operative bleeding, thromboembolic disease, vascular injury, deep periprosthetic joint infection, periprosthetic facture medial collateral ligament injury, and anesthesia-related complications [33]. Thus, GNA may be a safer alternative to KA should patients who receive this procedure not need subsequent KA for management of their pain. However, more research needs to be conducted specifically examining the safety profile and short-term and long-term complication rate of GNA given its novelty as a procedure.

4.3. Cost of GNA compared to KA

Another important consideration other than safety and efficacy of GNA is the cost burden of GNA for the individual and healthcare system. While the cost of GNA varies based on ablative technique utilized and geographic location, GNA can range from a few hundred dollars to 3000 dollars. Cost-effectiveness analyses done by Desai et al. on cooled GNRFA in the U.S showed an incremental cost of 18,773 and 84,392 dollars per quality-adjusted life-years (QALY) gained through cooled GNRFA when compared to intraarticular corticosteroid and HA respectively [34,35]. These numbers are below the willingness-to-pay threshold of 10,000 dollars per QALY gained illustrating that cooled GNRFA is a cost-effective option for tackling pain secondary to KOA when comparing to some of the available injection options. KA on the other hand has high upfront costs that range anywhere from 14,000 to 50,000 dollars depending on geographic location and procedure setting with the additional incurred expenses of post-operative rehabilitation that GNA does not entail. There are no studies directly comparing the cost-effectiveness of GNA and KA; however, the low percentage of patients in our cohort who eventually require KA after GNA demonstrates that GNA may provide individuals a lower cost option to effectively manage pain secondary to KOA and save them thousands of dollars.

4.4. Limitations

This study has several limitations. Because it relied on a large electronic database, individual patient records could not be reviewed to verify coding accuracy or ensure completeness of clinical data. Its retrospective design is also unable to establish causality. Reliance on procedural and diagnostic coding may introduce misclassification bias, as coding errors, variability in provider documentation, or differences in institutional billing practices could lead to inaccurate identification of patients, procedures, or comorbid conditions. Comorbidities precluding knee arthroplasty outside of what was excluded for within this study may not have been captured, which may have influenced the lower conversion rates noted in the current study. Furthermore, CPT code 64624 encompasses all GNA techniques – GNRFA (thermal, cooled, pulsed), cryoneurolysis, and chemical neurolysis. The relative efficacy of these approaches may differ, though comparative data remain limited. Adding onto this, there is wide heterogeneity in how these procedures are performed – whether a pre-procedural diagnostic block was performed, if fluoroscopy vs ultrasound guidance was used, differences in ablation targets, variations in probes used, etc. [10,11]. TriNetX does not capture laterality, and some arthroplasty cases identified after GNA may have involved the contralateral knee. There is existing literature discussing the use of pre-KA GNA for post-KA pain, which this study could not control for [36]. Variability in institutional follow-up intervals may also lead to under- or overestimation of conversion rates. Additionally, clinical parameters such as radiographic grade, baseline pain severity, and functional outcomes were not available. Finally, this study may be affected by selection bias, as patients who seek GNA may naturally be more inclined to avoid surgery in general.

Despite these limitations, the use of a large, multi-institutional dataset enhances the generalizability of these findings and provides a meaningful benchmark for future comparative studies.

4.5. Future directions

Future investigations should employ prospective, matched-cohort designs incorporating time-to-event analyses to better quantify how genicular nerve ablation (GNA) influences the timing and likelihood of conversion to knee arthroplasty. Long-term follow-up of patients undergoing specific GNA procedures compared with matched non-GNA controls would help clarify whether and to what extent GNA truly delays or prevents surgical intervention. Further research is also needed to standardize procedural protocols, including target selection, imaging guidance, energy modality, and the use of diagnostic blocks, to reduce variability and improve reproducibility across centers. In addition, comparative studies evaluating thermal, cooled, pulsed, cryoneurolytic, and chemical neurolysis techniques should assess not only relative efficacy and durability but also cost-effectiveness, to identify the most clinically and economically advantageous approach for managing chronic knee osteoarthritis pain.

While beyond the scope of the present study, similar investigations could be extended to other peripheral joints, such as the shoulder, where ablation techniques have shown promising analgesic outcomes [37]. Because current CPT coding lacks specificity for these joints, analogous analyses exploring the potential surgical-sparing or surgical-delaying effects of ablation in non-knee joints would be valuable to further define the role of peripheral neuroablative procedures in musculoskeletal pain management.

5. Conclusion

In this large real-world cohort, only about one in eight patients who underwent genicular nerve ablation later received knee arthroplasty. These findings suggest that GNA may offer meaningful symptom relief that may allow patients to defer surgery, supporting its role as a minimally invasive treatment option for symptomatic knee osteoarthritis. While this study has inherent limitations and cannot establish causality for whether GNA delays or prevents KA, it provides a valuable benchmark from which future prospective and comparative studies can build to clarify the durability, surgical-sparing potential, and cost implications of GNA.

Statement

During the preparation of this work the author(s) used ChatGPT-5 to generate tables and for limited language editing and phrasing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Data availability

Data are available from the corresponding author upon reasonable request and subject to TriNetX licensing restrictions.

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